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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

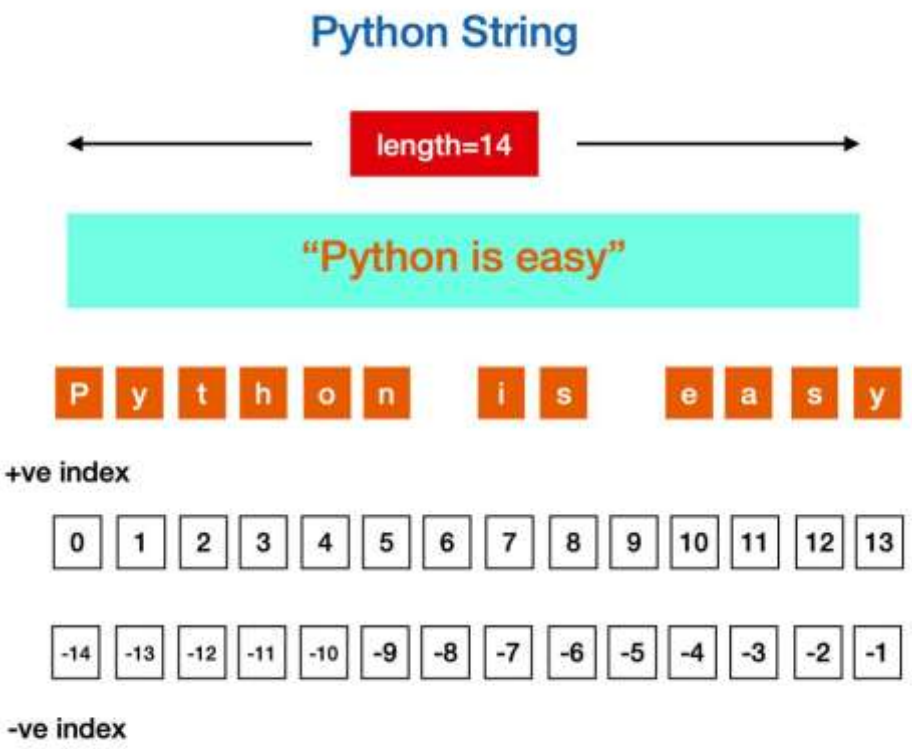
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 31: Strings in Python



What is a String?

A **String** is a sequence of **characters** (letters, numbers, symbols, or spaces).

In Python, strings are written inside:

- Double Quotes " "
- Single Quotes ' '

Example

```
name = "Rahul"

print(name)
```

Output

Rahul

Real-Life Example

Think about your:

- Name
- School Name
- City
- Favorite Subject

These are all **text**, and in Python, text is called a **String**.

Example:

```
Name = "Rahul"

School = "ABC High School"

City = "Hyderabad"
```

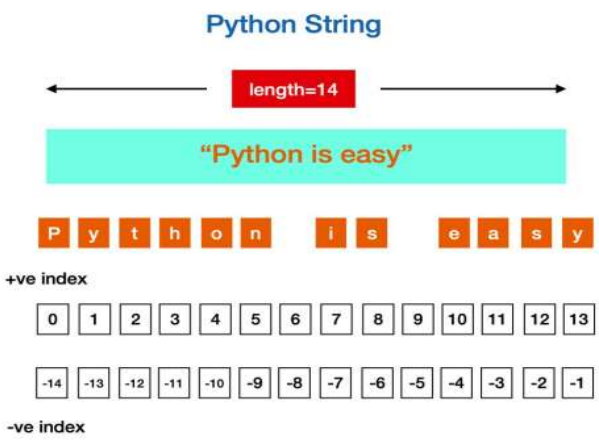
Features of Strings

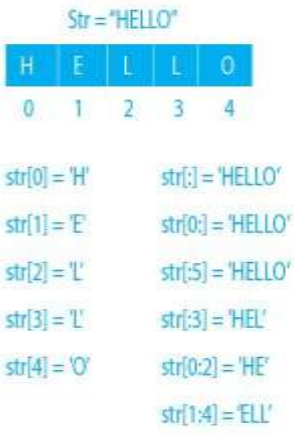
- Stores text data
- Written inside quotes
- Ordered collection of characters
- Supports indexing
- Supports slicing

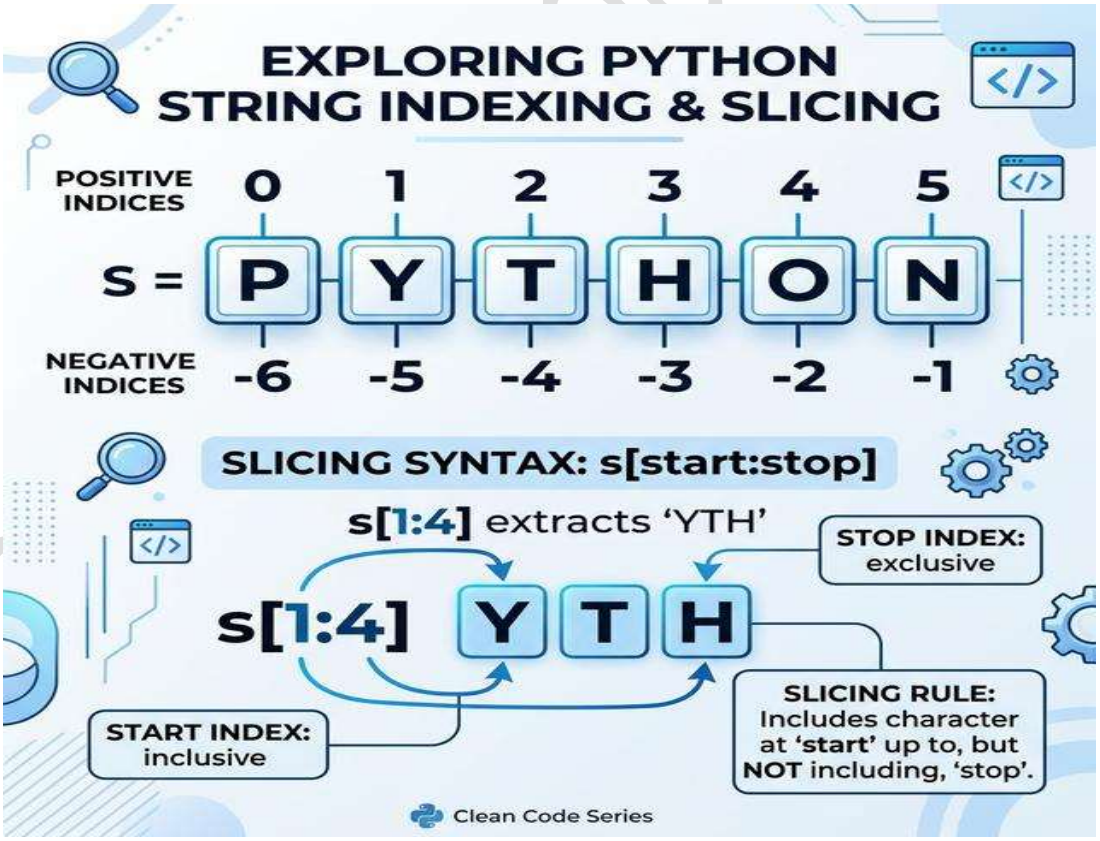
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- Immutable (cannot change individual characters)

What is a String?







String

"PYTHON"

P Y T H O N

0 1 2 3 4 5

Creating Strings

name = "Rahul"

city = 'Hyderabad'

print(name)

print(city)

Output

Rahul

Hyderabad

String Length

Use the len() function to count the number of characters.

name = "Python"

print(len(name))

Output

6

Accessing Characters (Indexing)

Each character has an **index number** starting from **0**.

word = "Python"

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print(word[0])

Output

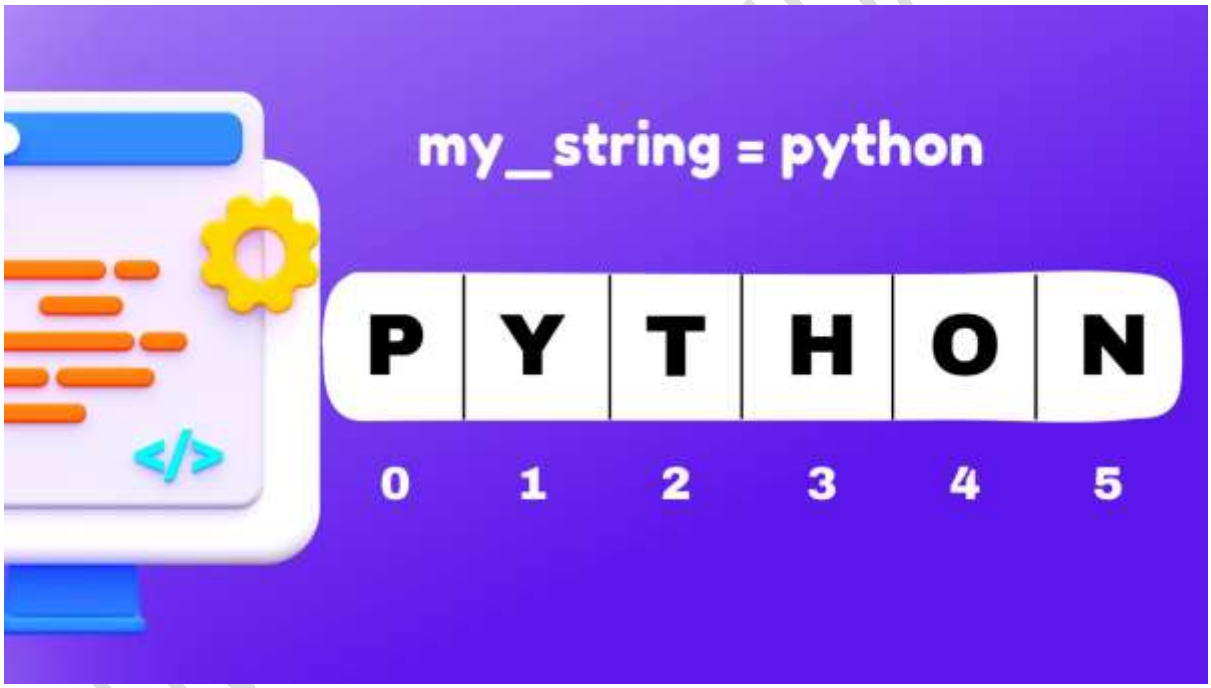
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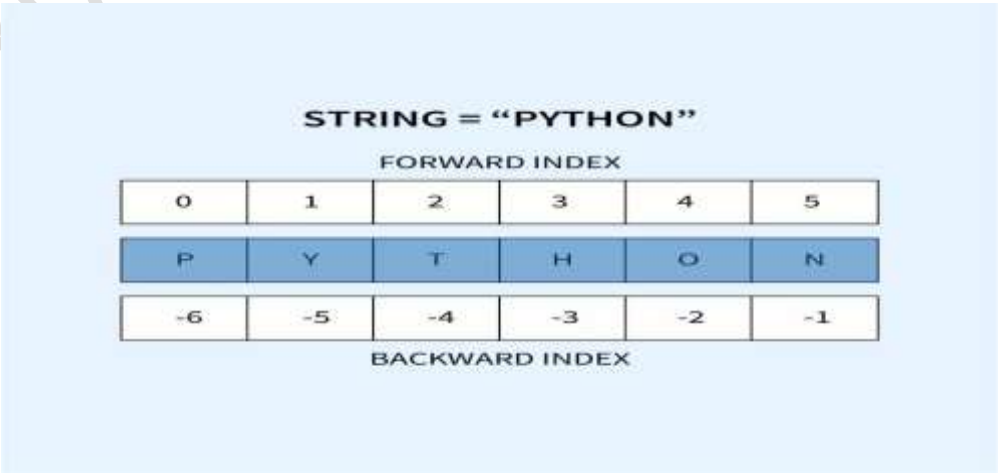
print(word[3])

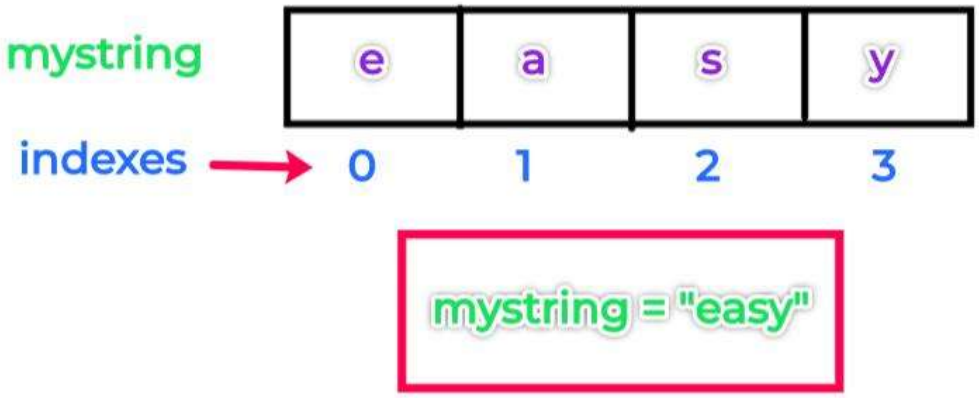
Output

h

String Indexing







Word

P Y T H O N

0 1 2 3 4 5

Negative Indexing

word = "Python"

print(word[-1])

Output

N

print(word[-2])

Output

O

String Slicing

word = "Python"

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```
print(word[0:3])
```

Output

```
Pyt
```

```
print(word[2:6])
```

Output

```
thon
```

Loop Through a String

```
name = "Python"

for letter in name:
   print(letter)
```

Output

```
P
y
t
h
o
n
```

Check if a Word Exists

```
text = "I love Python"

print("Python" in text)
```

Output

```
True
```

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Convert to Uppercase

```
name = "python"


print(name.upper())
```

Output

PYTHON

Convert to Lowercase

```
name = "PYTHON"


print(name.lower())
```

Output

python

Capitalize First Letter

```
name = "python"


print(name.capitalize())
```

Output

Python

Title Case

```
text = "welcome to python"


print(text.title())
```

Output

Welcome To Python

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Replace Words

```
text = "I like Java"


print(text.replace("Java", "Python"))
```

Output

```
I like Python
```

Remove Extra Spaces

```
text = "   Python   "


print(text.strip())
```

Output

```
Python
```

Split a String

```
text = "Apple Banana Mango"


print(text.split())
```

Output

```
['Apple', 'Banana', 'Mango']
```

Join Strings

```
fruits = ["Apple", "Banana", "Mango"]


print(", ".join(fruits))
```

Output

```
Apple, Banana, Mango
```

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String Concatenation

```
first = "Hello"

second = "World"

print(first + " " + second)
```

Output

```
Hello World
```

String Repetition

```
print("Hi " * 3)
```

Output

```
Hi Hi Hi
```

Find a Word

```
text = "Python Programming"

print(text.find("Programming"))
```

Output

```
7
```

Count Characters

```
text = "banana"

print(text.count("a"))
```

Output

```
3
```

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Starts With

```
text = "Python"


print(text.startswith("Py"))
```

Output

```
True
```

Ends With

```
text = "Python"


print(text.endswith("on"))
```

Output

```
True
```

String Formatting (f-String)

```
name = "Rahul"
age = 13


print(f"My name is {name}. I am {age} years old.")
```

Output

```
My name is Rahul. I am 13 years old.
```

Escape Characters

```
print("He said, \"Hello!\"")
```

Output

```
He said, "Hello!"
```

Strings are Immutable

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```
name = "Python"


name[0] = "J"
```

Output

TypeError

You cannot change a single character in a string.

Common String Methods

Method	Purpose
upper()	Convert to uppercase
lower()	Convert to lowercase
capitalize()	Capitalize first letter
title()	Capitalize every word
replace()	Replace text
strip()	Remove spaces
split()	Split into a list
join()	Join list into string
find()	Find text position
count()	Count occurrences
startswith()	Check beginning
endswith()	Check ending

Real-Life Examples

Student Name

```
student = "Anita"

print(student)
```

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School Name

school = "ABC Public School"

Greeting Message

message = "Good Morning!"

Practical Session

Practical Exercise 1

name = "Python"

print(name)

print(len(name))

Practical Exercise 2

city = "Hyderabad"

print(city.upper())

print(city.lower())

Practical Exercise 3

text = "I Love Python"

print(text.replace("Love", "Learn"))

Practical Exercise 4

word = "Computer"

for letter in word:

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```
print(letter)
```

Practical Exercise 5

```
first = "Good"

second = "Morning"

print(first + " " + second)
```

Mini Activity

Guess the Output

```
word = "Python"

print(word[2])
```

Answer

t

Guess the Output

```
print("Hello" * 2)
```

Answer

HelloHello

Classroom Activity

Create a program that:

1. Stores your name in a string.
2. Prints the first letter.
3. Prints the last letter.

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4. Prints the length of your name.
5. Converts your name to uppercase.
6. Converts your name to lowercase.

Quiz

Choose the Correct Answer

1. Which data type stores text?

A) Integer

B) Float

C) String

D) Boolean

✅ **Answer:** C

2. Strings are written inside

A) []

B) {}

C) Quotes (" " or ' ')

D) ()

✅ **Answer:** C

3. Which function gives the length of a string?

A) size()

B) count()

C) len()

D) find()

✅ **Answer:** C

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4. Which method converts text to uppercase?

A) lower()

B) upper()

C) capitalize()

D) title()

✅ **Answer:** B

5. What is the index of the first character in a string?

A) 1

B) -1

C) 0

D) 2

✅ **Answer:** C

Homework

1. Create a string with your name.
2. Print the first and last characters.
3. Find the length of your name.
4. Convert your name to uppercase.
5. Convert your name to lowercase.
6. Replace one word in a sentence.
7. Split a sentence into words.
8. Join a list of fruits into a single string.
9. Count how many times the letter "a" appears in "banana".
10. Write a program using an **f-string** to introduce yourself.

Key Points to Remember

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- A **String** is a sequence of characters used to store text.
- Strings are written inside **single (' ') or double (" ") quotes**.
- Strings support **indexing** and **slicing**.
- Use len() to find the number of characters.
- Strings are **immutable**, meaning individual characters cannot be changed.
- Python provides many useful string methods like upper(), lower(), replace(), split(), join(), find(), and count().
- Strings are widely used to store names, messages, addresses, and other text data.

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